Efficacy and Safety of Oncovex in Her2-low metastatic breast cancer: A Phase 3 Randomised, Double-Blind, Placebo-Controlled Trial

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Note: Entirely fictional study with invented drug and data, created for training. Do not use for clinical decisions.

Abstract

Background: Oncovex (OVX) is an intravenous antibody-drug conjugate under investigation for HER2-low metastatic breast cancer. **Methods:** In this multicentre trial, 512 adults were randomised 1:1 to Oncovex or placebo for 52 weeks, stratified by baseline severity and region. The primary endpoint was progression-free survival (months). **Results:** The primary endpoint was 9.6 with Oncovex versus 5.1 with placebo (p <0.001). Neutropenia grade 3+ occurred in 21.7% of Oncovex patients and 9.4% of placebo patients. **Conclusions:** Oncovex met its primary endpoint with a manageable safety profile in HER2-low metastatic breast cancer.

Methods

Eligible participants were aged 18 to 75 years with a confirmed diagnosis of HER2-low metastatic breast cancer and provided written informed consent. Exclusion criteria included pregnancy, severe hepatic impairment (Child-Pugh C), and concurrent use of strong CYP3A4 inducers. Randomisation used an interactive web-response system. The analysis followed the intention-to-treat principle with multiple imputation for missing data.

Results

Characteristic	Oncovex	Placebo
Participants (n)	256	256
Mean age (years)	56.4	53.1
Female (%)	44	43
Primary endpoint	9.6	5.1
p-value	<0.001	-
Neutropenia grade 3+ (%)	21.7	9.4
Discontinuation due to AE (%)	2.2	2.1

Serious adverse events were reported in 6.3% of the Oncovex group and 11.4% of the placebo group. There were 6 deaths in the Oncovex arm and 2 in the placebo arm, none judged related to study treatment.

Discussion

The effect size observed with Oncovex is clinically meaningful. Limitations include the 52-week duration, under-representation of patients aged over 75 years, and limited ethnic diversity. Long-term safety beyond one year requires an open-label extension, which is ongoing.

References

1. Synthetic reference list omitted. 2. Illustrative Guidelines on Management, 2025.